

Progress Report v7

Drug–Drug Interaction Prediction via Biomedical LLM Embeddings
and Verified Negative Sampling on OGB-ddi

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1 Research Question

The central research question of this project is:

Can the performance of the LLM + GCN framework on OGB-ddi be improved by (1) substituting biomedical-specific LLM embeddings (PubMedBERT, ChemBERTa) for general-purpose ones, and (2) replacing random negative sampling with clinically verified non-interacting pairs from CRESCENDDI — and if so, by how much compared to the state-of-the-art result of $AUC = 0.900$ reported by LLM-DDI [1]? Additionally, how do these configurations compare against classical classifiers (logistic regression, random forest, MLP)?

This framing makes three independent contributions evaluable:

1. The effect of **domain-specific embeddings** (PubMedBERT for text, ChemBERTa for molecular structure) versus general-purpose LLM embeddings.
2. The effect of **verified negative sampling** (CRESCENDDI) versus random unlabelled pairs.
3. A **classical baseline** comparison to quantify the contribution of graph structure over flat feature classification.

2 Background and Related Work

2.1 Drug–Drug Interactions

A drug–drug interaction (DDI) occurs when one drug influences the pharmacological activity of another, potentially causing adverse effects or reduced efficacy. As polypharmacy is increasingly common — particularly in elderly patients — automated DDI prediction is clinically critical. Exhaustive experimental testing of all drug pair combinations is infeasible, motivating computational approaches.

2.2 Key Methods

DeepDDI [5] (Ryu et al., 2018) uses structural similarity profiles as input to a deep neural network, achieving AUC-ROC of 0.926 on DrugBank.

KGNN [4] (Lin et al., 2020) proposes a knowledge graph neural network that captures topological neighbourhood information, achieving AUC-ROC of 0.991 on DrugBank V5.1.4.

LLM-DDI [1] (Li et al., 2026) integrates LLM-generated molecular embeddings (`text-embedding-ada-002`, dimension 1536) into a GNN-based framework evaluated on OGB datasets using 5-fold cross-validation. This is the primary baseline for this project.

Table 1: Published benchmark results from LLM-DDI [1] (Table II, p. 778).

Method	OGB-biokg				OGB-ddi			
	ACC	F1	AUC	AUPR	ACC	F1	AUC	AUPR
RF	0.910	0.911	0.972	0.973	0.730	0.733	0.804	0.787
KGNN	0.695	0.579	0.820	0.814	0.511	0.549	0.515	0.556
KG2ECapsule	0.937	0.938	0.982	0.971	0.735	0.752	0.794	0.744
LLM-DDI	0.964	0.964	0.994	0.994	0.822	0.824	0.900	0.883

The target for this project is $AUC = 0.900$ on OGB-ddi (Table 1).

2.3 Motivation for PubMedBERT and ChemBERTa

General-purpose BERT achieved near-random performance ($\text{AUROC} = 0.558$) in our preliminary BioSnap experiments (Section 6.3), directly motivating a domain-specific replacement. **PubMedBERT** was pretrained exclusively on PubMed abstracts and full biomedical articles, making it the natural choice for encoding drug text descriptions [10].

ChemBERTa was pretrained on 77 M SMILES strings from the ZINC database [11], giving it a vocabulary specifically suited to molecular representations. Its use for DDI prediction via GNN node initialisation has direct precedent in recent literature: PolyLLM [12] shows that ChemBERTa + GNN outperforms other embedding strategies for polypharmacy side effect prediction.

3 Datasets

3.1 Overview

Three datasets are used in this project, all ultimately derived from DrugBank [6] but differing in scope, preprocessing, and intended use.

Table 2: Dataset comparison. Node features are generated via drug descriptions (desc) and/or SMILES; OGB-ddi requires embedding generation from scratch.

Dataset	Drugs	Interactions	Sparsity	Mean degree	Components	Node features
BioSnap	1,514	48,514	4.2%	64.1	2	via desc/SMILES
DrugBank	1,706	191,808	13.2%	224.4	–	via desc/SMILES
OGB-ddi	4,267	1,067,911	11.7%	500.5	1	None (generated)

3.2 BioSnap

BioSnap is a collection of biomedical networks assembled by the Leskovec lab at Stanford [14]. The DDI subset (**ChCh-Miner**) is derived from DrugBank and provides a clean, citable benchmark graph. It is used as a development dataset in this project for validating the embedding pipeline before scaling to OGB-ddi.

Methodological improvement. The DDI-LLM notebook applies a joint SMILES + description filter, reducing BioSnap from 48,514 to $\sim 27,800$ edges even for description-only embeddings. In this project, the filter is applied *conditionally*: SMILES-based embeddings (ChemBERTa) require valid SMILES; description-based embeddings (PubMedBERT) use the full graph ($\sim 47,000$ edges), recovering a 69% increase in training data.

3.3 DrugBank

The DrugBank benchmark (1,706 drugs, 191,808 interactions) is a specific filtered snapshot of the full DrugBank database, used as a second validation dataset. Note that KGNN results were obtained on DrugBank V5.1.4 (a larger version), so direct comparison requires a caveat.

3.4 OGB-ddi

OGB-ddi is a standardised DDI benchmark from the Open Graph Benchmark [9] (Stanford, NeurIPS 2020), enabling direct comparison with published methods including LLM-DDI. It contains FDA-approved and experimental drugs connected by known interactions, using a **protein-target split** that ensures the test set contains drugs targeting different proteins than training drugs — a harder and more realistic evaluation than random splits.

Key properties: fully connected (single component, no isolated nodes); substantially denser than BioSnap (mean degree 500.5 vs. 64.1); no pre-computed node features (`graph.x = None`).

Split: train 1,067,911 / val 133,489 / test 133,489 edges.

4 Data Analysis

4.1 BioSnap Network Structure

Table 3: BioSnap network properties.

Property	Value	Interpretation
Largest connected component	1,510 / 1,514	4 peripheral drugs
Average degree	64.1	moderate connectivity
Degree assortativity	negative	hubs connect to low-degree nodes
Mean clustering coefficient	~ 0.25	bimodal distribution
Mean closeness centrality	0.40–0.45	2–3 hops between any two drugs
Degree vs. desc. length corr.	0.14	weak — embeddings add new info

The degree distribution is heavy-tailed: 38% of drugs have fewer than 15 interactions while a small number of hub drugs interact with over 400 others. The weak correlation ($r = 0.14$) between degree and description length confirms that graph structure and LLM embeddings capture complementary information.

4.2 OGB-ddi Network Structure

Table 4: OGB-ddi network properties.

Property	Value	Interpretation
Connected components	1	fully connected
Mean degree	500.5	8 $\times$ higher than BioSnap
Median degree	446.0	most drugs well-connected
Max degree	2,234	extreme hub drug
Min degree	1	no isolated nodes
Sparsity	11.7%	denser than BioSnap
Mean clustering coefficient	~ 0.516	2 $\times$ higher than BioSnap
Degree vs. desc. length corr.	0.174	weak, consistent with BioSnap

OGB-ddi is substantially denser than BioSnap. This density introduces more noise during GCN message passing, consistent with LLM-DDI finding OGB-ddi harder to model. The high clustering coefficient (0.516) indicates dense local community structure, which should support consistent GCN neighbourhood aggregation.

4.3 Description Quality — BioSnap and DrugBank

Two corruption types were identified: ; ; ; ; ($n = 6,512$ rows in the source file) and #NAME? ($n = 6$), both Excel artefacts. Only 14 drugs across both datasets are affected. Median description length is 90 characters (std = 245).

Table 5: Description and SMILES quality for BioSnap and DrugBank.

Feature	BioSnap		DrugBank	
	Count	%	Count	%
Valid descriptions	1,489/1,514	98.3%	1,695/1,706	99.4%
Corrupted (;;;)	8/1,514	0.5%	6/1,706	0.4%
Missing	17/1,514	1.1%	5/1,706	0.3%
Valid SMILES	1,332/1,514	88.0%	1,695/1,706	99.4%

4.4 OGB-ddi Coverage Analysis (new, this report)

A node-level coverage analysis was performed to determine the feasibility of generating biomedical LLM embeddings for all 4,267 OGB-ddi drug nodes. Drug descriptions (for PubMedBERT) and SMILES strings (for ChemBERTa) were retrieved from the DDI-LLM repository, which contains 15,235 drug entries covering the full DrugBank database. All OGB-ddi DrugBank IDs were matched via a left join.

Important correction from v6. The previous report incorrectly flagged 98.5% of OGB-ddi descriptions as corrupted due to the presence of ;;; sequences. These are in fact DrugBank *inline citation markers* (e.g. [L41539,L41569]), not Excel artefacts. This is structurally distinct from the BioSnap/DrugBank corruption pattern. No ;;; filtering is applied to OGB-ddi descriptions.

Table 6: OGB-ddi node-level coverage for embedding generation.

Category	Count	%
Total drugs	4,267	100.0%
Valid descriptions (PubMedBERT)	4,267	100.0%
Valid SMILES / RDKit (ChemBERTa)	3,543	83.0%
Both valid	3,543	83.0%
SMILES missing/invalid	724	17.0%
Neither valid	0	0.0%

Description length. Lengths range from 4 to 1,885 characters (mean 315.2, std 291.0, median 237). Notably, 1,012 drugs (23.7%) have descriptions shorter than 100 characters, which may produce lower-quality PubMedBERT embeddings due to limited textual context. This is flagged as a potential limitation.

SMILES fallback. The 724 drugs without valid SMILES are consistent with the expected proportion of biotech drugs (antibodies, proteins) for which SMILES do not exist. All 724 have valid descriptions, meaning no drug is entirely unrepresentable. These drugs will receive the mean ChemBERTa embedding computed over all valid entries.

5 Negative Sampling — CRESCENDDI

The original DDI-LLM pipeline samples negatives randomly from unlabelled drug pairs. This is methodologically weak since unlabelled pairs may be undiscovered positives. CRESCENDDI [8] provides 4,544 clinically verified non-interacting pairs. A mapping to DrugBank IDs was performed using the DrugBank vocabulary file (55,576 name-ID entries covering 19,853 drugs).

OGB-ddi has the highest CRESCENDDI coverage of the three datasets, as expected given its larger drug vocabulary. The 9 unmapped pairs correspond to the truncated drug name

Table 7: CRESCENDDI negative pair coverage across all three datasets. Drug overlap was computed explicitly for OGB-ddi; for BioSnap and DrugBank, usable pair counts were derived by checking whether both drugs in each pair appear in the respective dataset node list.

	BioSnap	DrugBank	OGB-ddi
Drugs in dataset	1,514	1,706	4,267
Unique CRESCENDDI drugs	434	434	434
CRESCENDDI drugs found in dataset	–	–	433 (99.8%)
Total CRESCENDDI pairs	4,544	4,544	4,544
Successfully mapped to DrugBank IDs		4,535 (99.8%)	
Usable pairs (both drugs in dataset)	4,306	4,078	4,531
Usable %	95.0%	89.9%	99.9%

mefenamate.

Class imbalance. The ratio of positive edges to CRESCENDDI negatives on OGB-ddi is approximately 220:1 (~1M vs. 4,531). Rather than oversampling or mixed negative sampling, the imbalance is handled via the `pos_weight` parameter of `torch.nn.BCEWithLogitsLoss` [13]. This upweights positive examples in the loss function to compensate for the scarcity of verified negatives, without repeating any data or modifying the training pipeline:

```
pos_weight = n_positives / n_crescenddi_negatives # ~236
criterion = BCEWithLogitsLoss(
    pos_weight=torch.tensor([pos_weight]).to(device)
)
```

This is numerically equivalent to having 236 copies of each positive example in the loss, rebalancing the gradient signal in a single line of code. The loss combines sigmoid and BCE in one numerically stable operation via the log-sum-exp trick.

Experimental design. Two conditions will be compared:

1. **Baseline:** train with random negative sampling (standard OGB-ddi protocol); evaluate on both random negatives and CRESCENDDI pairs at test time.
2. **CRESCENDDI:** train with 4,531 verified negatives + `pos_weight` rebalancing; evaluate on the same test sets.

Comparing these two conditions isolates the effect of negative sampling quality on both in-distribution performance (random negatives) and out-of-distribution generalisation (verified non-interactions).

6 Implementation

6.1 Environment

Google Colab Pro (H100 GPU) for model training and embedding generation. Local VS Code (Windows 11, Python 3.12, `.venv`) for data exploration and scripting. Packages cached to Google Drive to avoid reinstallation overhead. Key versions: `torch==2.11.0+cu130`, `torch_geometric==2.7.0`, `transformers==5.5.4`, `ogb==1.3.6`, `numpy==1.26.4`, `pandas==3.0.1`, `rdkit`, `scikit-learn`.

6.2 Bug Fix in DDI-LLM Pipeline

A bug was identified and fixed in the `LM()` embedding loading function: the original code dropped only 2 metadata columns for all embedding files, but the BERT description file contains

3 metadata columns (DrugBank ID, SMILES, and an index), causing SMILES strings to be incorrectly used as embedding features. The fix identifies metadata columns dynamically by dtype:

```
metadata_cols = [col for col in df.columns
                  if df[col].dtype == object]
features = df.drop(columns=metadata_cols)
```

6.3 Preliminary Results on BioSnap

Table 8: First results on BioSnap (single seed, lr = 0.0001).

Embedding	AUROC	AUPR
GPTDesc	0.974	0.934
BERTDesc	0.558	0.458

GPTDesc achieves 0.974 AUROC, consistent with published results. BERTDesc performs near-randomly (0.558), consistent with the original paper (max 0.750 across all learning rates).

The gap between GPT (0.974) and BERT (0.558) is striking given that both are general-purpose LLMs. It demonstrates that embedding *quality* matters enormously for this task — not merely the use of an LLM. BERT’s near-random performance suggests that drug descriptions require *biomedical* language understanding to be encoded meaningfully: a model pretrained on Wikipedia and books does not know what “thrombin inhibitor” or “cytochrome P450” means in the way a model pretrained on PubMed does. This gap directly motivates PubMedBERT as a domain-specific replacement, and is the primary scientific contribution of this project: testing whether domain-specific embeddings close this performance gap.

Multi-seed evaluation is pending.

6.4 Embedding Status

Table 9: Embeddings planned and their status.

Embedding	Input	Type	Dim	Status
BERT	Description	General LLM	768	Available (bug fixed)
GPT	Description	General LLM	1,536	Available, working
LLaMA/LLaMA2	Description	General LLM	–	Available
ChemBERTa	SMILES	Chemistry LLM	768	Available
Molformer	SMILES	Chemistry LLM	–	Available
PubMedBERT	Description	Biomedical LLM	768	Generated (saved to Drive)
Morgan FP	SMILES	Classical	1,024	To generate (RDKit)

7 Next Steps

Table 10: Planned tasks in priority order.

Task	Details
Confirm negative sampling strategy	Supervisor input on oversampling vs. evaluation-only
<i>PubMedBERT</i> embeddings [done]	Generated for all 4,267 OGB-ddi drugs (100% coverage); saved to Drive; sanity-checked via cosine similarity
Generate ChemBERTa embeddings	<code>seyonec/ChemBERTa-zinc-base-v1</code> ; run on Colab H100; mean-vector fallback for 724 drugs without SMILES
Fix conditional SMILES filter	Apply only for SMILES-based embeddings; recover $\sim 47,000$ BioSnap edges for description models
Classical baselines	LR, RF, MLP on concatenated/element-wise drug pair embeddings; evaluate AUROC and AUPR
GCN training on OGB-ddi	Plug PubMedBERT/ChemBERTa embeddings as node features; train link predictor using OGB train/val/test splits
Multi-seed evaluation	5 seeds per configuration; report mean $\pm$ std AUROC and AUPR; Wilcoxon signed-rank test
Error and overfitting analysis	Training loss and validation AUROC curves per epoch; early stopping (patience = 20); score distribution plots; separate CRESCENDDI hard negative evaluation

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